



PARIKALP 2026

ASTRO ALLIANCE

GENERAL INSTRUCTIONS

1. **Eligibility and Team Formation:** Participation is open to students competing in **teams of 2-4 members**. Interdisciplinary teams are permitted, provided all members to meet the registration criteria.
2. **Programming Environment:** Implementations must be built **strictly using Python**. Participants are required to deploy their pipelines utilizing standard python libraries and frameworks. The use of **Google Colab or Jupyter Notebook (.ipynb)** environments is highly preferred for structural evaluation.
3. **Submission Deliverables:** Teams must submit the final .ipynb file of their solution notebook. The source code must be well-commented to clearly explain the logic.
4. **Project Report & Visualizations:** **Alongside the code file, each team must submit a comprehensive project report (PDF)** documenting their end-to-end pipeline strategy, methodology, and results. This report must explicitly feature analytical visualizations, including model comparison bar graphs, error metric charts, and feature importance plots, complete with proper explanations and descriptions.
5. **Defensive Evaluation Protocols:** Deliverables will be manually evaluated based on pipeline clarity, structural feasibility, algorithmic innovation, and the mathematical viability of the drawn conclusions.



OVERVIEW

THE 2084 SURVIVAL CHALLENGE

The Crisis

The year is 2084. **Astro Alliance's** manned exploration shuttle, ***PARIKALP***, was sent into deep space to study a newly discovered cluster of planets. Disaster struck when an unexpected gravitational pull dragged the shuttle off course, trapping it in the powerful orbit of a mysterious exoplanet. The main engines are severely damaged. The crew has only enough backup fuel for one single emergency launch. There are no second chances.

The Problem

To break free from the planet's gravity and escape into safe space, the crew must fire the engines to hit the exact Escape Velocity ($v_e = \sqrt{\frac{2GM}{R}}$).

Too slow: The shuttle will crash back down onto the alien surface.

Too fast: The strained backup fuel tanks will explode.

To make matters worse, violent cosmic storms have wrecked the shuttle computer network. The sensor data showing the planet's mass, radius, and gravity are corrupted, filled with heavy noise, errors, and missing values. Because raw data is such a mess, standard physics equations cannot be used directly. **In the chaos following the gravitational anomaly, a miscommunication between mission scientists led to one of the distance parameters being logged in two different units.** Unless this hidden inconsistency is detected and corrected, any prediction could prove fatal.

Your Directive

You are the lead Space Data Scientist in the mission presented on board. Your mission is to build a reliable predictive pipeline using Python with a very limited amount of time. By training machine learning models on clean historical planet data taken from the Earth itself, your pipeline must clean the noisy sensor streams, reconstruct the missing planetary features, and accurately predict the exact escape velocity.



MARKING SCHEME

Section No	Sub-section No	Section / Sub-section	Marks
1	-	Exploratory Data Analysis (EDA)	25
	1.1	Data Inspection	6
	1.2	Statistical Inference	8
	1.3	Feature Selection, Dimensionality Reduction & Encoding	6
	1.4	Visualization	5
2	-	Numerical Interpretation & Mathematical Analysis	14
	2.1	Feature Engineering	10
	2.2	Additional Features	4
3	-	Radius and Mass Prediction	13
4	-	Model Interpretation	13
5	-	Conclusion	5
		Total	70

NOTE: The dataset contains missing and anomalous values due to observational errors. Participants are expected to identify physics-based inconsistencies through visualization and apply appropriate mathematical relationships for imputation and correction.

1. Exploratory Data Analysis (EDA)

(25 points)

1.1 Data Inspection:

(6 points)

- Inspect the dataset and determine the data types of all features.



- Generate and interpret descriptive statistics for all numerical variables.
- Identify features that have missing values.
- Identify the numerical and categorical features of the dataset to use for further analysis.
- Handle missing values using appropriate imputation techniques.

1.2 Statistical Inference:

(8 points)

- Plot the distribution of numerical features to assess the skewness of the data.
- Normalize the data if needed.
- Detect and handle outliers.
- Perform noise detection and analyze its impact.
- Explore the relationship between features using scatter plots or **correlation** matrices.
- Analyze the distribution of target variables, including skewness, outliers, and potential prediction challenges.

1.3 Feature Selection, Dimensionality Reduction & Encoding:

(6 points)

- Dropping unnecessary columns.
- Checking for data leakage columns.
- Appropriate encoding of categorical features with justification
- Feature selection.
- Application and interpretation of dimensionality reduction techniques.

1.4 Visualization:

(5 points)

- Identification of physics law violations through visualization.
- Formulation and application of appropriate mathematical relationships for imputation

2. Numerical Interpretation and Mathematical Analysis

(14 points)

2.1 Feature Engineering:

(10 points)

- Planetary Feature Engineering
- Orbital Feature Analysis
- Stellar Feature Engineering
- Photometric & Spectral Features
- Innovative Feature Creation
 - Novel features beyond provided suggestions
 - Scientific/mathematical justification
 - Demonstrable impact on model performance



2.2 Additional Features:

(4 points)

- Creation of additional features using domain knowledge, statistical insights, or feature interactions to improve predictive modeling.

3. Radius and mass prediction:

(13 points)

- Model Selection and Implementation
- Multi-output Regression Models
- Cross-validation
- Hyperparameter Tuning
- Model Evaluation (MAE, RMSE, R^2)
- Prediction Visualization (Actual vs Predicted)
- **Prediction of Escape Velocity using the predicted mass and radius.**

4. Model Interpretation:

(13 points)

- Regression Metrics (MAE, RMSE, R^2)
- Cross Validation Results
- Hyperparameter Optimization
- Error Analysis & Residual Plots
- Comparison of Multiple Models (at least 2)
- Accuracy Analysis
- Bias-Variance Analysis

5. Conclusion:

(5 points)

- Key findings
- Future scope
- Limitation



RULES AND REGULATIONS FOR SOLUTION EVALUATION

- All code submitted must be the original work of the participating team.
- The solution should be reproducible, with clear documentation of preprocessing, feature engineering, and model development steps.
- Proper justification must be provided for all assumptions, imputations, and feature engineering techniques used.
- Models will be evaluated based on accuracy, robustness, interpretability, and scientific validity.
- Any form of target leakage, data leakage, or use of information derived directly from target variables will result in penalties.
- Additional marks may be awarded for innovative approaches, novel features, and proper code comments.
- A clear presentation of methodology, results, and conclusions is mandatory.
- Teams must clearly explain model selection, hyperparameter tuning, and evaluation-metrics.
- **Use of LLMs or any AI-assisted tools is strictly prohibited.** AI detection will be used to verify all submissions.
- Judges reserve the right to question any result, assumption, or methodology used in the solution.
- In case of a tie, preference will be given to solutions demonstrating better scientific reasoning, optimization, interpretability, and innovation.
- The decision of the evaluation committee shall be final and binding.



- Astro Alliance reserves the right to amend any rules, regulations, or evaluation criteria as deemed necessary.